

Build the Coupled-Coil Instrument

A low-voltage pulse and resonance platform

ChatGPT edition — the course synthesis,
inside the learner boundary

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

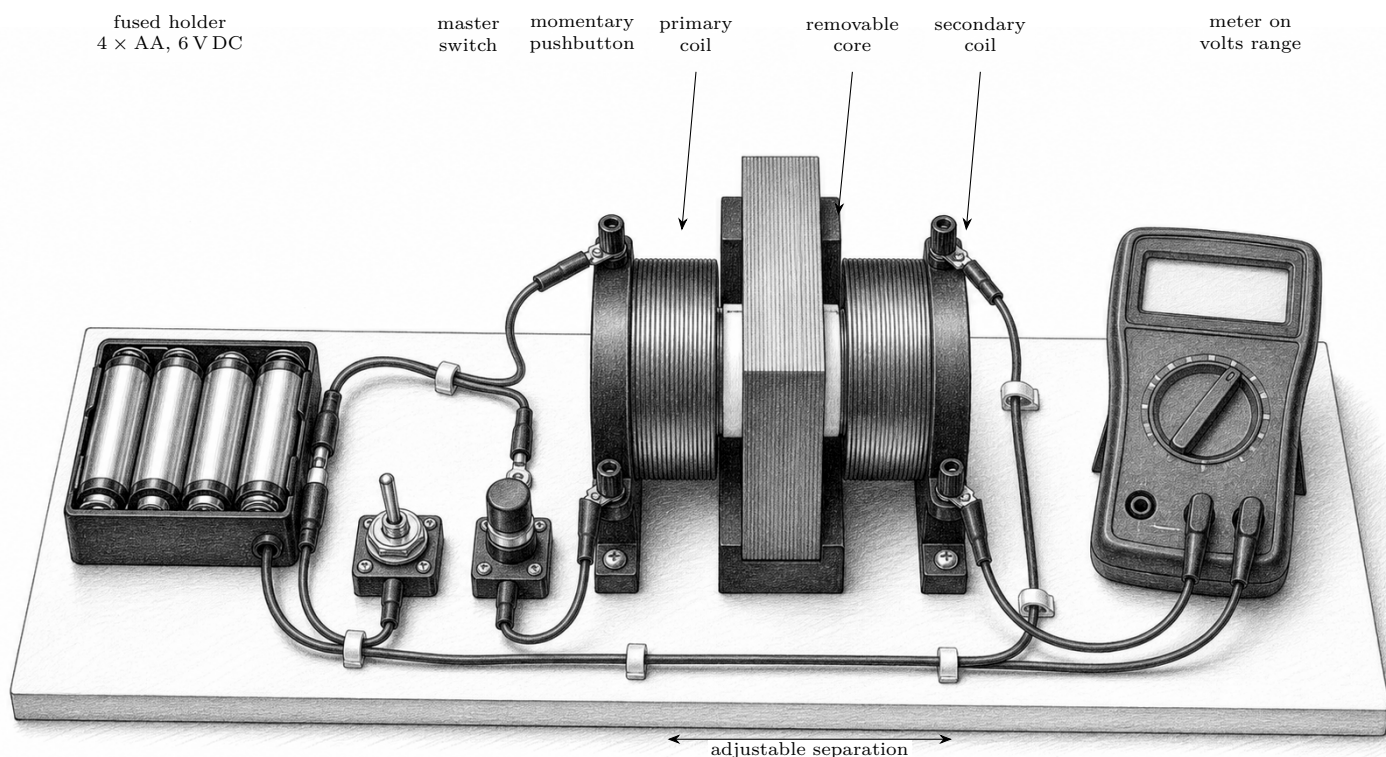
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- 2–4 AA cells in a fused, switched holder
- Two cardboard bobbins, insulated magnet wire, and a removable ferrite or laminated-iron core
- Momentary pushbutton, multimeter, ruler, masking tape, and fine sandpaper
- Cardboard base, insulated jumpers, and cable ties
- Optional: two LEDs wired antiparallel, each with its own 1 k Ω resistor

Predict first

Which change should most increase the receiver indication: aligning the coils, adding the core, or holding DC continuously?



Read the finished instrument before building

The primary is the only coil connected to the battery. The secondary is an isolated receiver connected only to the voltmeter or the optional resistor-limited LED pair. The core slides through both bobbins; it is not an electrical conductor in either circuit. Every change happens with the master switch off and the battery holder disconnected.

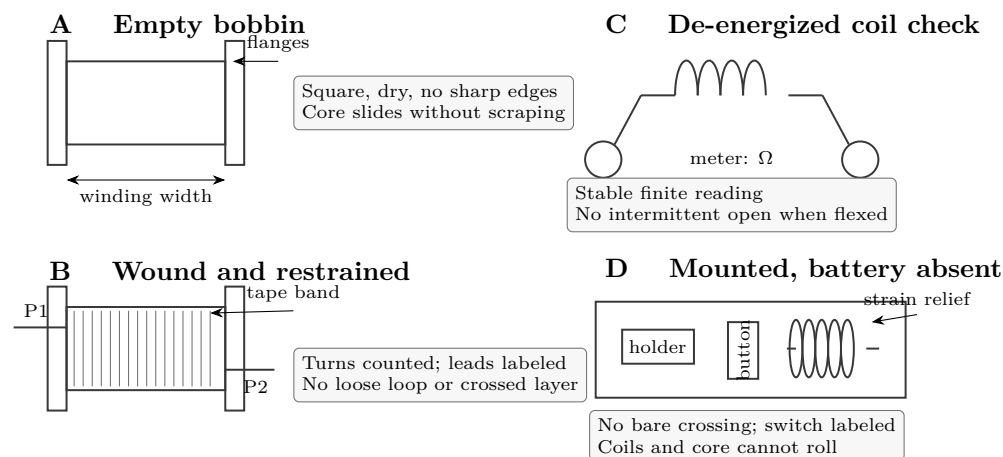
Investigate

Gate 0: freeze the design on paper

Choose one primary and one secondary bobbin that fit the same removable core. Plan 100–200 neat turns per coil; use the same turn count first so geometry, not ratio, is the first variable. Draw the lead names P1/P2 and S1/S2 before winding.

Predict first

Where could enamelled wire become bare? What prevents a loose turn, a pulled lead, or a battery lead from crossing another terminal? Point to each control on the drawing before touching materials.



Gate 1: make and measure each coil

1. With the battery holder absent, inspect each bobbin. Round or tape every edge that could scrape enamel. Measure winding width and core clearance.
2. Leave a 10 cm lead and tape it down as P1. Wind the primary in one direction, laying turns beside one another. Count aloud in groups of ten and mark each group on paper. Never wind on a powered circuit.
3. Tape the final layer without hiding the two lead ends. Leave 10 cm for P2. Repeat for S1/S2 on the second bobbin.
4. The adult removes enamel only from the last 1 cm of each lead. With the meter on ohms and no battery present, record R_P and R_S . Gently flex one lead at a time: the reading must remain finite and stable.

Predict first

If the meter says OL, is the coil a conductor? If it reads nearly zero, what must be ruled out before connecting cells? Explain why a visual turn count cannot replace the resistance test.

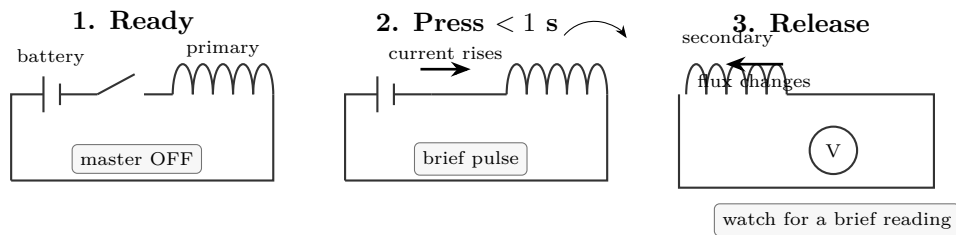
Gate 2: mount the powerless instrument

Fasten the holder, master switch, pushbutton, bobbins, and meter leads to the base. Add strain relief before making electrical joints. Slide the core through both bobbins and confirm that it can be removed without touching a terminal. Wire the primary path as holder–fuse–master switch–pushbutton–primary–holder. Connect the secondary only to the voltmeter. The adult traces both paths with a finger, then uses continuity mode to verify that OFF is open and a pressed button conducts only when the master switch is ON.

Predict first

Can current travel from either secondary lead to either primary lead? The required answer is no continuity. If there is continuity, stop: what contact, stray strand, or mistaken connection could have joined the isolated circuits?

Gate	Measure	Pass evidence	If it does not pass
Bobbin	core clearance; winding width	slides freely; edges smooth	reshape or replace before winding
Primary	turns; R_P ; flex test	count recorded; finite, stable R_P	expose/clean leads; rewind if open
Secondary	turns; R_S ; flex test	count recorded; finite, stable R_S	find broken lead or rewind
Mounting	exposed copper; pull test	none; every part restrained	insulate and add strain relief
Power path	battery voltage; switch state	expected holder voltage; OFF is open	adult corrects wiring; do not pulse
Pulse	peak indication, 3 trials	repeatable response on press/release	use diagnostic path below

Gate 3: verify one controlled pulse

Set the meter to a low DC-voltage range on the secondary. Start with the coils aligned, touching, and sharing the core. The adult installs 2 AA cells, reads the holder voltage, and turns the master switch on. The learner presses the button for less than one second and releases it. Record the largest visible secondary indication on press and on release. Switch OFF and disconnect the holder. Repeat three times only after confirming that the coil and holder remain at room temperature.

Predict first

Why may the meter briefly indicate one sign on press and the other on release? Why is holding the button down not a stronger induction test? What observation would make you stop immediately?

Gate 4: change one thing

Choose exactly one: remove the core, add a measured gap, rotate one coil, or change the cell count within 2–4 AA cells. Keep turns, meter range, pulse time, and lead orientation fixed. Make three trials at each setting. Between trials: master OFF, holder disconnected, touch-check for warmth, then reposition.

Diagnostic path: no repeatable indication

- Power absent?** Disconnect the coils. Adult verifies holder voltage, fuse, master switch, and pushbutton continuity. Reconnect only after the path passes.
- Primary open?** With cells removed, remeasure P1–P2 while gently flexing leads. OL or jumping values means repair or rewind.
- Receiver open or mis-set?** Remeasure S1–S2, confirm leads are in the meter COM/V jacks, and confirm volts rather than amps or ohms.
- Coupling weak?** Return to zero gap, coaxial bobbins, and the core fully inserted. Recheck one brief pulse.
- Instrument too slow?** Reverse the secondary leads and observe both press and release. A display may miss a short transient; that is a limit of the measurement, not permission to raise voltage or make sparks.

Stop if a fuse opens twice, a coil warms, insulation is damaged, or any result is unexplained. Do not bypass the fuse and do not substitute mains, a large battery, a capacitor, an ignition coil, or a spark gap.

Explain from the evidence

This instrument combines current-produced field, changing flux, mutual induction, and geometry. The secondary responds while primary current changes—at press and release—not simply because DC exists. Alignment, small gap, a shared magnetic core, and turn count affect linked flux. The instrument deliberately does not make sparks and is not a Tesla coil. A repeatable low-energy pulse and an honest record provide better evidence than an uncontrolled high-voltage spectacle.

Change one thing

Make a calibration plot of peak receiver indication versus separation: 0, 5, 10, 20, and 40 mm, three trials each. Add range bars from the smallest to largest trial. Then complete this acceptance record:

Primary	turns _____	$R_P =$ _____ Ω
Secondary	turns _____	$R_S =$ _____ Ω
Supply / fuse	_____	
Baseline geometry	core _____	gap _____ mm
Three baseline peaks	_____	
Stop checks	no warmth [] no odor [] insulation intact []	
Adult inspection	initials _____	date _____

Photograph or redraw the accepted wiring and label every energy-transfer stage.

Evidence record	
Prediction	_____
One change	_____
Observation	_____
Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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